AST101 – Astronomy of the Solar System at Suffolk County Community College, NY

Catalog Description:
Introduction to fundamental aspects of planetary science. Topics include historical development of astronomy; basic concepts of celestial coordinates and motions; properties and individual characteristics of planets and their moons, asteroids, comets and meteoroids; and origin and evolution of the solar system. Students also learn to identify celestial objects (constellations, prominent stars, planets, etc.) utilizing planetarium, telescopes and unaided eye. Occasional evening observations required. Note: Fulfills SUNY-GE Natural Sciences. (3 hrs. lecture, 2 hrs. laboratory) Prerequisite: MAT007 (Algebra I) or equivalent. Offered on: A-E-G / 4 cr. hrs.

Learning Outcomes:

Upon completion of this course, students will be able to:
1. Use measurements and astronomical data to describe the solar system.
2. Summarize how the scientific method applies to astronomy.
3. Use the horizon system to locate prominent stars and constellations.
4. Compare the fundamental properties of planets to Earth and each other.
5. Evaluate the Sun's influence on the planets of the solar system.
6. Relate the properties of the solar system to the solar nebula theory.
7. Discuss recent developments in the field of astronomy.

Because of the basic algebra math prerequisite, AST101 is transferable as a general science elective to astronomy-major programs at 4-year institutions, and not as a major-specific course (there are few exceptions). Most AST101 students take the class to satisfy a science requirement to graduate with an associate degree. For most students, it is the first and only science class they take in college.

Sample of instructional schedule choices by Glenda Denicolo (modality: in person):

W	Date	HW # due	Activity	Textbook Chapter/Section
1	Wed Sep 3		Introduction to AST101; *pre-assessment to evaluate the course* **Lab 1**: Planetarium with https://stellarium-web.org/; **Planetarium room**	
2	Mon Sep 8		Brief overview of astronomy	1
	Wed Sep 10	1	Scales, the universe in powers of ten, units, constellations, horizon system **Lab 2**: Overview of mathematical and scientific tools	2.1, 4.1, App C, D
3	Mon Sep 15		Patterns and cycles in the sky, seasons	2.1, 4.1-4.2
	Wed Sep 17	2	Basic properties of light, electromagnetic spectrum **Lab 3**: Basic coordinates and seasons	5
4	Mon Sep 22		Atomic transitions, types of astronomical spectra	5
	Wed Sep 24		Doppler effect, astronomical instruments **Lab 4**: The horizon system and analyzing the rotating sky	5, 6.2, 6.5
5	Mon Sep 29	3	**Test 1**	
	Wed Oct 1		Early models of the solar system, Kepler's laws **Lab 5**: Planetary orbits – Part 1	2
6	Mon Oct 6		Scientific method, Newton's laws	3
	Wed Oct 8	4	Gravity, angular momentum, energy **Lab 6**: Planetary orbits – Part 2	3
7	Mon Oct 13		The Moon	9.1-9.4
	Wed Oct 15	5	The Moon **Lab 7**: Planetary orbits – Part 3	9.1-9.4, 4.5-4.7
8	Mon Oct 20		The Sun	15, 16
	Wed Oct 22		The Sun **Lab 8**: Lunar phases – Part 1	15, 16
9	Mon Oct 27		The Sun	15, 16
	Wed Oct 29	6	Formation of solar system	7.1-7.2, 7.4, 14.3

			Lab 9: Lunar phases – Part 2	
10	Mon Nov 3	7	**Test 2**	
	Wed Nov 5		Planetary geology **Lab 10**: The Sun	7.1-7.2, 8.2, 10.1
11	Mon Nov 10		Planetary geology	7.1-7.2, 8.2, 10.1
	Wed Nov 12		Planetary atmospheres **Lab 11**: Craters on planetary surfaces**; Planetarium room**	8.3, 10.3, 10.5-10.6, 11.3
12	Mon Nov 17		Mercury	9.5
	Wed Nov 19	8	Venus, Mars **Lab 12**: Solar system properties	10.1-10.4
13	Mon Nov 24		Mars, meteorites	10.4-10.6, 14.1-14.2
	Wed Nov 26		*No class – Thanksgiving recess*	
14	Mon Dec 1		Asteroids, comets	13
	Wed Dec 3	9	Jovian planets **Lab 13**: Atmospheric retention	11
15	Mon Dec 8		Jovian planets, their moons, and the Kuiper Belt	11, 12
	Wed Dec 10		*Post-assessment to evaluate the course.* **Extra credit presentations; Planetarium Test**	
16	Mon Dec 15	10	**Test 3 (1:15 min)**	
	Wed Dec 17		**Optional cumulative Final Test (1:15 min)**	

Textbook: https://openstax.org/details/books/astronomy-2e/
Labs and homework: https://www.pivotinteractives.com/ (keyword AST1SCCC for the labs)

AST102 – Astronomy of Stars and Galaxies at Suffolk County Community College, NY

Catalog Description:
Introduction to fundamental aspects of universe beyond our solar system. Topics include properties of electromagnetic radiation and its relation to study of celestial objects; structure, classification and evolution of stars, nebulae, star clusters, galaxies, and material between stars. Age, origin and evolution of universe studied in terms of modern cosmology. Occasional evening observations required. Note: Fulfills SUNY-GE Natural Sciences. (3 hrs. lecture, 2 hrs. laboratory.) Prerequisite: MAT007 (Algebra I) or equivalent. Offered on: A-E-G / 4 cr. hrs.

Learning Outcomes:

Upon completion of this course, students will be able to:

1. Use measurements and astronomical data to describe the universe.
2. Summarize how the scientific method applies to astronomy.
3. Explain how the properties of light are used to obtain information about the universe.
4. Compare the fundamental properties of stars to the Sun and each other.
5. Describe the structure of the Sun.
6. Trace the evolution of stars from birth to death.
7. Compare the fundamental properties of galaxies to the Milky Way and each other.
8. Describe the past and future evolution the universe.
9. Discuss recent developments in the field of astronomy.

Because of the basic algebra math prerequisite, AST102 is transferable as a general science elective to astronomy-major programs at 4-year institutions, and not as a major-specific course (there are few exceptions). Most AST102 students take the class to satisfy a science requirement to graduate with an associate degree. For most students, it is the first and only science class they take in college. Students can choose to take AST101 or AST102, no matter the order.

Sample of instructional schedule choices by Glenda Denicolo (modality: in person):

W	Date	HW # due	Activity	Textbook Chapter/Section
1	Tue Sep 2		Introduction to AST102; *pre-assessment to evaluate the course*	
	Thu Sep 4		Brief overview of astronomy **Lab 1**: Planetarium with https://stellarium-web.org/; **Planetarium room**	1
2	Tue Sep 9		Scales, the universe in powers of ten, units, constellations, horizon system	2.1, 4.1, App C, D
	Thu Sep 11	1	Patterns and cycles in the sky, seasons **Lab 2**: Overview of mathematical and scientific tools	4.1-4.2, App C, D
3	Tue Sep 16		Basic properties of light, electromagnetic spectrum	5
	Thu Sep 18	2	Atomic transitions, types of astronomical spectra **Lab 3**: Basic coordinates and seasons	5
4	Tue Sep 23		Doppler effect, astronomical instruments	5, 6
	Thu Sep 25		Newton's laws, Law of gravitation **Lab 4**: Gravity and orbits – Part 1	3
5	Tue Sep 30	3	**Test 1**	
	Thu Oct 2		The Sun (structure, composition, solar cycle) **Lab 5**: Gravity and orbits – Part 2	15
6	Tue Oct 7		The Sun (nuclear fusion)	16
	Thu Oct 9	4	Analyzing star light (brightness, color, spectra) **Lab 6**: The Sun	17
7	Tue Oct 14		Stellar spectra	17
	Thu Oct 16	5	Surveying stars (mass, diameter) **Lab 7**: What is this star made of?	18
8	Tue Oct 21		Surveying stars (variable stars, star clusters)	19, 22
	Thu Oct 23	6	Surveying stars (formation of stars) **Lab 8**: The HR diagram – Part 1	20, 21
9	Tue Oct 28		Life cycle of stars (cycle for low-mass stars)	22
	Thu Oct 30		Life cycle of stars (cycle for high-mass stars) **Lab 9**: The HR diagram – Part 2	22
10	Tue Nov 4	7	**Test 2**	
	Thu Nov 6		Stellar graveyard (white dwarfs, neutron stars, pulsars, binary systems, blackholes, gamma-ray bursts) **Lab 10**: Distance to the stars (brightness and luminosity)	23, 24
11	*Tue Nov 11*		*No class – Veteran's day*	
	Thu Nov 13		The Milky Way (characteristics) **Lab 11:** Cosmic distance ladder	25
12	Tue Nov 18		The Milky Way (stellar populations, formation, its center)	25
	Thu Nov 20	8	Galaxy clusters; distance measurements **Lab 12**: Expanding universe; **Planetarium room**	26
13	Tue Nov 25		Galaxy evolution, Quasars, Active Galactic Nuclei (AGNs)	28
	Thu Nov 27		*No class – Thanksgiving recess*	
14	Tue Dec 2		Evidence for dark matter	27
	Thu Dec 4	9	Dark matter and large-scale structure formation; dark energy **Lab 13**: Doppler effect	28, 29
15	Tue Dec 9		The Big Bang; Cosmic Microwave Background (CMB) radiation	29
	Thu Dec 11		*Post-assessment to evaluate the course.* **Extra credit presentations; Planetarium Test**	
16	Tue Dec 16	10	**Test 3 (1:15 min)**	
	Thu Dec 18		**Optional cumulative Final Test (1:15 min)**	

Textbook: https://openstax.org/details/books/astronomy-2e/
Labs and homework: https://www.pivotinteractives.com/ (keyword AST2SCCC for the labs)